Jaeyoung Cha

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Personal Profile

I am a third-year Ph.D. student in the Optimization & Machine Learning (OptiML) Laboratory at Kim Jaechul Graduate School of AI, Korea Advanced Institute of Science and Technology (KAIST AI), where I am fortunate to be advised by Prof. Chulhee Yun. Previously, I completed my M.Sc. in AI at KAIST and B.E. in Cyber Defense (minor in Mathematics) at Korea University.

Research Interests

I study the theoretical foundations of modern deep learning and design algorithms grounded in rigorous analysis. My work [C1, C4] establishes lower bounds on convergence rates for without-replacement SGD. Another line of work [C2, C3] develops and theoretically justifies an algorithm that strengthens length generalization in Transformers. My recent work [W1, U1] develops sampling strategies to improve inference efficiency in speculative decoding with tree-structured drafts. I am also investigating the differences between low-rank adaptation (LoRA) and full fine-tuning from a theoretical perspective.

Education

Korea Advanced Institute of Science and Technology — *Seoul, South Korea*
Ph.D. in Artificial Intelligence — *Mar. 2024 – present*
- Advisor: Prof. Chulhee Yun

Korea Advanced Institute of Science and Technology — *Seoul, South Korea*
M.Sc. in Artificial Intelligence — *Mar. 2022 – Feb. 2024*
- Advisor: Prof. Chulhee Yun
- Thesis: *Optimal Convergence Rate of Without-Replacement SGD*
- GPA: 4.23/4.3

Korea University — *Seoul, South Korea*
B.E. in Cyber Defense — *Mar. 2018 – Feb. 2022*
- Minor in Mathematics
- Overall GPA: 4.38/4.5 (Ranked 1st/30), Minor GPA: 4.5/4.5

Seoul Science High School — *Seoul, South Korea*
High School — *Mar. 2015 – Feb. 2018*
- GPA: 4.15/4.3

Publications

Published

*: Equal Contribution.

[C4] Yujun Kim*, **Jaeyoung Cha***, Chulhee Yun. Incremental Gradient Descent with Small Epoch Counts is Surprisingly Slow on Ill-Conditioned Problems. **ICML 2025**. OpenReview

[C3] Hanseul Cho*, **Jaeyoung Cha***, Srinadh Bhojanapalli, Chulhee Yun. Arithmetic Transformers Can Length-Generalize in Both Operand Length and Count. **ICLR 2025**. OpenReview

[C2] Hanseul Cho*, **Jaeyoung Cha***, Pranjal Awasthi, Srinadh Bhojanapalli, Anupam Gupta, Chulhee Yun. Position Coupling: Improving Length Generalization of Arithmetic Transformers Using Task Structure. **NeurIPS 2024**. OpenReview

- Short version in ICML 2024 Workshop on Long-Context Foundation Models (LCFM).

[C1] **Jaeyoung Cha**, Jaewook Lee, Chulhee Yun. Tighter Lower Bounds for Shuffling SGD: Random Permutations and Beyond. **ICML 2023 Oral** (Top 155/6,538=**2.37%**). OpenReview

[W3] **Jaeyoung Cha***, Yujun Kim*, Chulhee Yun. The Phase Transition in Random Reshuffling: Tight Rates under Strong Convexity. **NeurIPS 2026 Workshop OPT: Optimization for Machine Learning**. OpenReview

[W2] Minseop Shin, **Jaeyoung Cha**, Chulhee Yun. In-Context Learning of Hidden Markov Models by Loop Transformers. **NeurIPS 2026 Workshop on AI for Stochastic Dynamics: From Theoretical Foundations to Scientific Applications**. OpenReview

[W1] **Jaeyoung Cha***, Hanseul Cho*, Chulhee Yun. Layer Verification Accelerates Speculative Tree Decoding. **ICML 2026 Workshop AdaptFM: Resource-Adaptive Foundation Model Inference**. OpenReview

Domestic Conferences

[DC1] **Jaeyoung Cha***, Junghyun Lee*, Chulhee Yun. Cumulative Distribution Regret Minimization with Max Quantile Threshold in Multi-Armed Bandits. **Korea Computer Congress (KCC) 2026**.

Under Review

[U1] **Jaeyoung Cha***, Hanseul Cho*, Chulhee Yun. Beyond Local Acceptance: Subtree-Weighted Sampling for Speculative Tree Decoding. *Under Review.* 2026.

Awards

2026 **Best Paper Award**, Korea Computer Congress (KCC 2026)

2026 **Gold Reviewer**, ICML 2026

2024 **Top Reviewer**, NeurIPS 2024

2023 **Samsung Electronics DS Division Outstanding Collaboration Paper Award**, Samsung

2023 **Oral Presentation**, ICML 2023

2020 – 2021 **National Science & Technology Scholarship**, Korea Student Aid Foundation

2018 – 2021 **Military Scholarship**, Ministry of National Defense, Republic of Korea

Talks

Aug 2024. **SNU-KAIST AI/ML Theory Workshop, Gangneung, South Korea**. *Tight Bounds and Convergence Analysis of Shuffling SGD for Finite-Sum Minimization.*

Jun 2023. **KAIST SAARC Workshop, Busan, South Korea**. *Tighter Lower Bounds for Shuffling SGD: Random Permutations and Beyond.*

Teaching Experience

Fall 26 **AI505 Optimization for AI**, Teaching Assistant
Spring 26 **AI616 Deep Learning Theory**, Teaching Assistant
Fall 25 **AI505 Optimization for AI**, Teaching Assistant
Spring 25 **AI709 Advanced Deep Learning Theory**, Teaching Assistant

Academic Service

Conference Reviewer. ICLR (2025–2027), ICML (2025–2026), NeurIPS (2023–2024, 2026), AISTATS (2027)

Journal Reviewer. TMLR (2025), JOTA (2025–2026)

Skills

Languages. Korean (*native*), English (*proficient*)

Programming & Tools. Python, PyTorch, LaTeX